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Antifragility in IT-projects:
How risks and uncertainties can be dealt with in a complex world

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Abstract:

The managing of risks in projects is becoming increasingly harder in the modern world because of increased complexity. IT projects are subject to extreme cost overruns and failures because of risks and dependencies that impact the successful completion of them. The purpose of this study is to study how the handling of risks and uncertainties in IT projects relate to the concept of antifragility. Six project members from Swedish companies have been interviewed to find out how they think about risks and how they handle them. The data was analysed with a framework for antifragility in projects to see how the antifragile concepts relate to the risk management in IT projects. The results show that there exists a lot of risks related to IT projects, mainly concerning lack of communication, flexibility and ability to adapt, and that antifragile concepts can be used to better look at the handling of risks and uncertainties in IT projects.

Keywords:

Risk, Risk management, Uncertainty, Antifragility, IT projects, Complexity

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1 Introduction

This chapter will give an introduction to the problem of failed IT-projects and the increased risk and uncertainty of today's world that impact organizations and projects. The chapter also gives an introduction to the concept of antifragility, which describes the characteristics of a system that can deal with risks and uncertainties, and why this might give us a valuable perspective on how risks could be handled. Finally the purpose of the study and research question is presented.

1.1 Background

Designing software and systems have increasingly gotten more complex, where people have become more dependent on the correctness and safety of these systems, and where these systems change dynamically throughout the life cycle (Weyns et al., 2010).

There have been some cases of failed IT-projects in Sweden that have been given a lot of attention in the last couple of years, because of their failure and extreme costs for taxpayers. The latest of these is the *Millenium* system which is a journaling system that was made to replace several systems for the healthcare industry in the Swedish region Västra Götaland. The project has been a huge disaster in many ways and has gotten a lot of criticism for its lack of usability, bugs, security problems and costs. During the three days the system was in use, 42 planned surgical operations were cancelled due to complications with the system.
The cost of the project so far has been over 6 billion SEK and even though the project was paused in November 2024, the costs for the hundreds of consultants still working on the project is around 40 million SEK every month (Sande, 2024; Syrén, 2025; *Västra Götalandsregionen*, 2025).

Other examples include a digital learning platform for Stockholm city which was scrapped after costing 1 billion SEK (Olsson, 2023) and a train management system being ended after eight years, costing over 1 billion SEK (Dagens Nyheter, 2025).

When evaluating project performance, the extent to which a project is completed within the budget constraints, constitutes a huge part (Flyvbjerg et al., 2022). Studies have shown that large IT-projects run on average 45% over budget, and some of them even 200% over budget (Flyvbjerg et al., 2022). One reason for this is the interdependencies in technical components and system modules that can cause delays and errors in many parts of the development process (Flyvbjerg et al., 2022). One important factor is also the failure of recognizing the impact of the social system on the technical system. Bostrom and Heinen (1977) argues that computer-related technology is neutral, and the outcome and success of the system depends solely on the decisions that are made on how it should be used, which are influenced by the designer’s knowledge, skill, values and assumptions about people and organizations (Bostrom & Heinen, 1977).

Modern organizations are increasingly experiencing unexpected disruptions, in part of an accumulation of factors influencing our world, such as an increase in dependency on technology, political instability, economic crisis, regulatory changes and more (Ramezani & Camarinha-Matos, 2020).

1.2 Problem

One of the greatest problems for project-based organizations is the managing of unexpected disruptions, which are events or changes in the environment which can impact the realization of project value (Usher et al., 2024).

The environment where the system operates has a variety of stressors that can impact the functions and successfulness of the system (Johnson & Gheorghe, 2013). Stressors are the forces from outside the specified operating conditions that may move a system from their intended state towards an unintended state, where the state describes the correctness of the system's functions (Johnson & Gheorghe, 2013). Stressors increase the organization's need to adapt and realign itself (Botjes et al., 2021).

Requirements, environmental assumptions and usage profiles continuously change in the dynamic context, which means an increased amount of uncertainty when designing system properties. The changes in the context of these systems are hard to predict when the system is initially built, and leads to a higher uncertainty because the initial assumptions may no longer be true (Grassi et al., 2024).

A system is a collection of interconnected parts, where the parts perform their specific roles which gives the overall system a greater functionality. This works through sharing information, feedback loops and interactions within the parts of the system. The elements of the system can be composed of software, hardware, people, policies and nearly anything that contributes to the desired output of the system (Johnson & Gheorghe, 2013).

Fragility describes a system which easily can fail when exposed to stress. There are two conditions for fragility, firstly that the system experiences harm when exposed to disruptions, and secondly the absence of any gain from disruptions (Usher et al., 2024).

Every system has a fragility threshold that, when crossed due to stressors, will break down the system. The size and interconnectedness in a system will impact its fragility, where a larger system, proportionally will have a lower fragility threshold and where interconnected systems will share the fragility threshold of its weakest component (Usher et al., 2024).

Robustness describes the ability of the system (or system characteristics) to remain in a desired state when exposed to varying amounts of stress. It describes an increased tolerance and lack of sensitivity to stress (Johnson & Gheorghe, 2013). All systems are expected to perform and create value in a range of different circumstances and challenges (Usher et al., 2024). Robust systems are however focused on resistance and withstanding disruptions, and are limited when adapting to changing environments (Usher et al., 2024).

How systems respond to stressors can be characterized on a continuum from fragility to robustness to antifragility, where the fragile system degrades with stress, the robust system is unchanged and the antifragile system improves (Taleb, 2012).

Antifragility describes a system which gets better (up to a point) when exposed to stress. Regular exposures to small doses of stress can strengthen the system and protect it from bigger doses of stress in the future (taleb) (Taleb, 2012). Antifragile systems use disruptions as a catalyst for growth and use them to create value (Usher et al., 2024).

1.3 Purpose

The purpose is to study how people in IT-projects can manage problems and how this relates to the concept of antifragility. There exists a research gap on how IT-based projects can handle the disruptions that they are faced with. Antifragility is a general concept and the aim of this study is to study how this concept relates to more tangible efforts that IT organizations use in their projects. Antifragility could in itself be evaluated based on how uncertainties and risks are handled, therefore an antifragility project is the description of a project that handles risks and uncertainties well.
The main motivation for the study is to study a new perspective to understand how risks and uncertainties can be dealt with in IT-projects, as the traditional risk management strategies have shown to be lacking when dealing with an increased complexity.

Operationally this means that the thesis will study how risks are being managed and how problems are being dealt with in IT-projects, since the arising of problems is the realization of risks.

Research question: How does the handling of risks and uncertainties in IT-projects relate to the concept of antifragility?

2 Review of the literature

This chapter will go through the literature that is related to risk handling and antifragility. The chapter begins with explaining what a system, project and organization is, to contextualize the uncertainties and complexities that encompasses this world. Thereafter, general risk and project management is presented. Lastly, a description of the antifragility scale is presented, to give an understanding of the different risk-handling capabilities of systems that exist and how they differ from each other.

2.1 IT- Projects and riskmanagement

2.1.1 IT-projects

An IT-system is composed of multiple technological components with different subsystems and applications, that together creates a comprehensive system (Flyvbjerg et al., 2022).
The word "system" comes from the ancient greek word "σύστημᾰ" (sústēmă) which means a whole made of several parts, or members ("Σύστημα," 2024). This understanding can be a useful abstraction that can be used to study different phenomena at different scales. It can also be used to understand how elements interact to generate behaviour at a system level (Gershenson, 2020).
A socio-technical system is made up of two interacting systems, the social system and the technical system (Bostrom & Heinen, 1977). The technical system transforms input into outputs with processes, tasks and the needed technology to do so. The social system is focused on the attributes of people and their relationships and authority structures. Together these systems produce outputs in the overall work system and the design of the system needs to take both of these systems in consideration (Bostrom & Heinen, 1977).

Systems provide vital needs by providing capabilities that are not possible by discrete components. These capabilities can include human activities, physical products, information products, mechanical functions or logical decisions (Johnson & Gheorghe, 2013).

2.1.2 IT-project problems

Flyvbjerg et al. (2022) studied cost overruns in IT projects and found that the probability distribution of cost overruns in IT-projects feature extreme values in the tail of the distribution that follows a power-law pattern, meaning that disastrous IT projects are not that rare as they sometimes are interpreted as. They mean that normal distributions cannot be used to understand the risk of extreme cost overruns that exist in IT-projects (Flyvbjerg et al., 2022).

Software projects often also take more time to be finished and provide less functionality and quality than expected (Barros et al., 2002). Many of such outcomes can be related to inadequate project management, communication problems, wrong allocation of team skills, insufficient training and the managers inability to predict and adjust the project behaviour (Barros et al., 2002).

Research shows that there exists difficulties building IT systems because of inherent interdependencies between components, leading to delays and errors during different phases of a project (Flyvbjerg et al., 2022). Building IT systems with interconnected technical components requires substantial efforts on making these components work together and where failures in one component can impact other interconnected components (Flyvbjerg et al., 2022).

In the environment where systems operate lies a variety of stressors, which can compromise the functionality of the system and hinder the completion of its objectives (Johnson & Gheorghe, 2013). Systems may however initially perform as expected during a certain lifespan but when they fail, it is often difficult to correct problems, especially when systems get increasingly complex (Jones, 2014).

Solutions for the uncertainty in complex systems isn't entirely technical, it must include social, political and economic considerations, therefore blending knowledge bases from different communities (Jones, 2014).
One of the biggest challenges for project-based organizations is the managing of sudden disruption, that quickly occurs and that is particularly impactful during the implementation phase (Usher et al., 2024).

2.1.3 Uncertainty

There exists two main interpretations of uncertainty: ontological randomness and epistemological randomness (Derbyshire & Wright, 2014). The ontological perspective says that prediction is impossible because the causes leading to outcomes are entirely based upon chance itself. The epistemological perspective says that there exists a deterministic relationship between the causes and their outcomes, here the uncertainties comes from the difficulty in identifying, unravelling and measuring causes (Derbyshire & Wright, 2014). Taleb (2008) means that there does not exist a functional difference between these perspectives if we, in practice, can't locate and understand relevant causes (Taleb, 2008).

Organizations have long been exposed and challenged by uncertainties, where the unpredictability and impact of the uncertainties pose significant threats (Adobor, 2025).

Uncertainties give rise to risks, which can lead to losses, and project managers are managing the uncertainties that can affect a project (Houston et al., 2001). These uncertainties can for example be if a key person leaves the project, or if there is a lack of accurate information on customer expectations (Houston et al., 2001).

2.1.4 Risk and uncertainty management in IT-projects

Project management requires the development of strategies that handle various risks and disruptions both internal and external, regardless of their source (Usher et al., 2024). Ropponen (2000) defines a risk as a state or property of a development task or environment that if ignored, will increase the likelihood of project failure (Ropponen & Lyytinen, 2000).

In product development, the risks are generally assumed to be consequences of a cause-effect relationship, however as the financial crisis showed, some risks have extreme consequences that can not be understood from a cause-effect perspective (Kennon et al., 2015). These risks are described by Taleb (2008) as “black swans”, which are rare events that have extreme impacts and which only are predictable retrospectively.

Traditional risk management strategies are based upon risk identification and quantification, which rely on past data and an ability to understand and a means of characterizing it (Houston et al., 2001; Kennon et al., 2015). Current methods for handling risks and uncertainties focus on minimizing or eliminating uncertainty from the start, this includes scenario planning and forecasting in strategic planning (Adobor, 2025). Organizations have also handled uncertainties for example by optimizing processes, using predictive models and employing collaborative approaches (Adobor, 2025).

These methods have been shown to work in moderately uncertain environments, however with the increasingly turbulent, complex, and dynamic organizational landscapes, these may show to be inadequate (Adobor, 2025). They also have shown to be unable to handle low-probability events with high impact (Ramezani & Camarinha-Matos, 2020).

2.2 Antifragility

The concept of fragility and antifragility was first described by Nassim Taleb in his book *Antifragile: Things that gain from disorder* from 2012. Fragility describes how a system gets worse from the variability of its environment past a certain threshold, and antifragility describes when a system gets better from the variability in its environment (Taleb, 2012; Taleb & Douady, 2012).
To classify if a system is fragile or antifragile, it is not needed to know the history and statistics of a system, only to evaluate if it accelerates towards harm or benefit with exposure to variability. Past data can however reveal valuable information about risks of certain processes, something that has broken before can be considered breakable, but something which hasn't broken cannot be considered unbreakable (Taleb, 2014).

The “variability” in a system’s environment describes the volatility or stress that a system can be exposed to. Taleb (2012) defines a “disorder cluster” which encompasses this definition: i) uncertainty, ii) variability, iii) imperfect, incomplete knowledge, iv) chance, v) chaos, vi) volatility, vii) disorder, viii) entropy, ix) time, x) the unknown, xi) randomness, xii) turmoil, xiii) stressor, xiv) error,xv) dispersion of outcomes (Taleb, 2012).

2.2.1 Fragility

Fragile systems are particularly exposed to harm from “tail” events, meaning that large deviations have a much larger impact than the class of events around the mean and center of a

distribution. Systems that are exposed to tail events also suffer a lot from uncertainty, because when systems such as buildings, bridges, airplanes, or nuclear plants are being built, they are made to be robust to certain levels of variability and stress but they may fail if this level is exceeded, meaning that they are fragile to uncertainty about the distribution of stressors, which rises the probability of failure or collapse of the system (Taleb, 2012; Taleb & Douady, 2012). A system can be considered to be more robust when it can handle a broader range of stress and a system's lack of sensitivity (or increased tolerance) to stress also leads it to being considered more robust (Johnson & Gheorghe, 2013).

The fragility of a system is gauged based on the response to the volatility of a stressor that affects the system. The fragility of a coffee cup is determined based on the response of a source of randomness or stress, and the fragility of a house is determined based on the response to the distribution of earthquakes etc. (Taleb & Douady, 2012).

Fragility is a relevant concept when understanding systems that exposed to extreme and unexpected events (also called "*black swans"*), because a fragile organization is harmed significantly more by a single, high impact event, than by several events of small or intermediate size (Derbyshire & Wright, 2014; Taleb, 2012). Fragile systems are necessarily immune to the cumulative effect of small shocks and disproportionately more impacted by larger ones. The coffee cup, for example, is more likely to experience large harm rather than small, and it can often only be described as healthy or broken, with nothing in between (Taleb, 2014; Taleb & Douady, 2012).

2.2.2 Antifragility

In contrast to fragile systems, antifragile systems experience benefits rather than harm from the disorder cluster (Taleb & Douady, 2012). The antifragile property is part of everything that has changed and survived with time, such as cultures, evolution, technical innovations or ideas. The antifragile is the difference between the organic and complex, and physical objects. The antifragile loves uncertainty and volatility, and certainly errors, as errors contain information that feeds the system with necessary details about its environment. Understanding antifragility is crucial to understanding fragility, as they both exist on a spectrum, we cannot improve health before reducing disease, or increasing wealth before decreasing losses (Taleb, 2012).

However, fragility in sub-components of an antifragile system is needed for the improvement of the whole, fragile elements need to exist so that they can disappear and be replaced by better (Gorgeon, 2015). Taleb (2012) gives an example of that when you starve yourself of food, the first organisms to break down are the bad ones which get recycled by your body, which is an evolutionary process that kills the weakest for fitness (Taleb, 2012). Survival pressures in an organism plays a role in its overall improvement under external stress, which is why fragility in sub-elements are needed (Taleb, 2012).

Adobor (2025) gives an example of fragility in business, where Blockbuster, a giant video rental company, failed to adapt to a shift in the market landscape, where video rental services were digitized and increasingly moved onto the internet, coupled with ineffective leadership

and debt leading to their bankruptcy. Fragile systems are easily broken or destroyed when exposed to unexpected stress, so they are heavily reliant on that things follow the plan (Gorgeon, 2015).

Fragile systems respond non-linearly to stress, it can repeatedly withstand small variations over time, and the accumulated effect of this variation won't affect the system, it is when the stressor passes a certain threshold the system breaks (Gorgeon, 2015; Taleb, 2012).

3 Theoretical framework

This chapter will present the analytical lens of the study, and what components of antifragility that will be looked at when analysing the data. The analytical lens is based on a framework by Usher et al. (2024) which presents antifragile concepts in the context of projects in general. These concepts build on what Nassim Taleb (2012) has described in his book "*Antifragile: things that gain from disorder*". These concepts are also expanded on by other authors, which is also presented in this chapter.

Usher et al. (2024) proposes a framework for how projects can embrace uncertainties. The described framework is based upon antifragile concepts that stem from Nassim Taleb (2012). These strategies are made in mind with that antifragile strategies need to be preparatory rather than preventive, as preventive strategies would need to have data on the causes and frequency of a disruption, which in a complex environment is not possible (Usher et al., 2024). Preparatory strategies, however, focus on strengthening the system so that the potential disruptions don't have as much impact (Usher et al., 2024).

3.1 Simplicity

Simplicity strategies focus on removing unnecessary elements, simplifying the project scope, choosing simple rather than complex planning strategies, promoting flexibility in project management, etc. (Usher et al., 2024). Simplicity strategies can help to remove inherent fragility, leading to a less complex system, making it easier for projects being able to respond and adapt to changing conditions (Usher et al., 2024).

Simplifications however need to be adopted by the organization as a whole and understood and discussed by business stakeholders, as simplifications in only the IS function may lead to rigidity in the organization as a whole (Gorgeon, 2015).

3.1.1 Subtraction

Subtraction is a strategy that aims to remove unnecessary elements or reduce complexity from the system to make it more resistant to shocks. By eliminating unnecessary parts of a system, the system gets more able to adapt to change (Usher et al., 2024).

For projects, this can mean that you practice scope management, and remove elements or features that don't align with the core objectives of the project as well as simplifying the project's scope (Usher et al., 2024). This is also reinforced by agile methodologies, where you focus on high value tasks and minimizes the focus on unnecessary tasks (Usher et al., 2024).

3.1.2. Less is more

The less is more rule is a strategy where organizations use flexible rules to guide their key processes and decisions, instead of having rigid plans (Usher et al., 2024). Taleb (2012) describes that the management of complex systems need to have simple rules and regulations. This is related to the "Ockham's razor" problem solving strategy, which builds on simple explanations rather than complex ones (Mardaras et al., 2021).

By having simple heuristics, project teams can respond flexibly and quickly when faced with new information or challenges, as well as the possibility to make more informed judgements under the conditions of uncertainty (Usher et al., 2024).

3.2. Development strategies

Development strategies aim to enhance the project adaptability, they provide flexibility in project planning and help with handling inherent uncertainties that projects face (Usher et al., 2024).

3.2.1. Building optionality

Optionality describes the system's availability of options. In the face of uncertainty, options create the possibility, but not the obligation to choose the best way going forward (Taleb, 2012). This builds into the fact that socio-technical systems are complex and the behaviour and possible disruptions cannot be completely understood or predicted. Optionality thus makes the system able to take advantage of the positive side of uncertainty (Taleb, 2012). Optionality is a non-causal approach to prepare for the future, it is a shield for our lack of capability of predicting the causation of events and their outcomes (Derbyshire & Wright, 2014).

One part of optionality is redundancy. Redundancy exists in natural systems as a central way to manage risk and unexpected events, for example the human body has two kidneys and extra capacity in many things, such as the lungs or neural system (Taleb, 2012). Redundancy is not only a defensive strategy to handle unexpected disruptions, it also is opportunistic, giving us the capability to act in the face of a disruption, making the most out of the situation (Taleb, 2012). Redundancy makes things simpler, making us not worry about trying to predict future unexpected events.

> "*If you have extra cash in the bank (in addition to stockpiles of tradable goods such as Spam and hummus and gold bars in the basement), you don't need to know with precision which event will cause potential difficulties*" (Taleb, 2012. p. 135)

There has however been an increase in efforts trying to remove redundancies in contemporary organizations in order to streamline and make the system more efficient (Derbyshire &

Wright, 2014). However, this increases the coupling between systems and takes away the buffer that defends the system against unexpected events, reducing our capability to take advantage of disruptions (Derbyshire & Wright, 2014).
For organizations, redundancy could mean that the IT-support is decentralized to each business unit, or that employees are trained for several jobs and that they can replace others if needed (Gorgeon, 2015). In projects, this could mean that there is enough slack in deadlines for individual tasks, so that a delay in some task doesn't jeopardize the completion of the whole project (Derbyshire & Wright, 2014).

Another part is trial and error which works by trying and learning from mistakes in the real world, it gives insight into what works and what doesn't (Gorgeon, 2015). Failures provide useful information that we can use to make better decisions to learn from (Taleb, 2012).
It is achieved by searching out information and adjusting thereafter. Trial and error includes small-scale experimentation which tests the market and can drive innovation, before committing to larger-scale investments (Gorgeon, 2015; Derbyshire & Wright, 2014). This builds on the fact that many important discoveries and inventions have happened by accident and come as a result from experimentation and tinkering (Taleb, 2012). This exists both in the actual invention or discovery, but also in its area of application (which can be much harder), for example many of the drugs we use today were initially designed to treat something else (Taleb, 2012).

Trial and error can often be seen in software development, where teams can create simple prototypes with little time and resources to gather information about the feasibility and customer interest (Usher et al., 2024).

3.2.2. Modularity

Modularity is a combination of simplicity and optionality, which can work by breaking down a complex system into subsystems and has long been used as a strategy to reduce complexity in the field of project management (Ramezani & Camarinha-Matos, 2020; Usher et al., 2024). Modularity can increase the positive convexity, evolvability and transferability to the project, by allowing an option to expand if it proves to be beneficial (Usher et al., 2024). This also allows projects to easier find and isolate weaknesses in certain parts of the system.

Gershenshon (2020) discusses the different types of control for different engineered systems depending on the complexity in the system and how the causality works. Here he proposes that self-organizing control, as opposed to centralized control, can deal better with high-complexity environments and where the causality is multiscale (both bottom-up and top-down). This relates to modularity as it is a division of system properties. Centralized control is though easier to implement and understand and is useful for low problems with low complexity and variety (Gershenson, 2020).

Gorgeon (2015) explains that most organizations today have a more decentralized form of IT governance, where organizations share the power and accountability to a wider range of external stakeholders, which may reduce the fragility in the information systems (Gorgeon, 2015).

One example of modularity in projects is the adoption of agile methodologies, such as Scrum or Kanban, which work by breaking down activities in smaller iterations and increments, which allows projects to adapt faster to changing customer needs or external changes (Usher et al., 2024).

Modularity also facilitates agility and flexibility by synchronizing the individual capabilities of subsystems, processes, etc. (Ramezani & Camarinha-Matos, 2020). Without modularity, the system becomes very tightly coupled, which makes the system more fragile, potentially leading to increasing amplitude and likelihood of failures (Johnson & Gheorghe, 2013).

3.3 Risk and opportunity strategies

Risk and opportunity strategies aim to prepare the project when faced with uncertainties, organizations can adapt faster to changes and handle disruptions better when actively managing their opportunities (Usher et al., 2024).

3.3.1. Barbell strategy

The barbell strategy suggests investing the majority of assets safely, i.e. options that are not exposed to negative disruptions and then taking risks with the remaining assets, thus still allowing for large-scale positive gains (Ramezani & Camarinha-Matos, 2020; Taleb, 2012). It is an avoidance of the middle, combining the maximally safe and maximally speculative options, removing the potential for risk for ruin (Taleb, 2012). This strategy does not intend to reduce the uncertainty, it however aims to reduce the potential negative impact of uncertainty, while still leaving the open for rare potential positive outcomes (Usher et al., 2024; Taleb, 2012) .

For personal finance, this can mean that you invest 90% of your funds in non-volatile options and 10% in very risky investments that have a small chance of large positive outcomes, thus limiting the potential losses to only 10% of the total investment while the potential gains are greater (Taleb, 2012).

Another example is where Google decided that 20% of their employees' time will be focused on experimentation as a way to drive innovation (Zwieback, 2014). The potential downside here was limited to 20% of the employees time, but the upside was much bigger, as has been shown with products such as Gmail, Google Maps and Adsense that all stemmed from this program (Zwieback, 2014).

For projects this can mean that, based on priorities of the project, you allocate the majority of resources to critical components to make sure that the minimum requirements are met to avoid project termination, and then using the remaining resources for innovative solutions for potentially highly beneficial parts, such as speed to market (Usher et al., 2024).

3.3.2. Skin in the game

Skin in the game is the idea that people that make decisions that affect others, also will have to take the consequences of their decisions, either good or bad (Taleb, 2018). The lack of skin in the game is a problem in organizations, where few people can make risky decisions, while personally escaping the potentially negative effects of its outcomes, transferring the fragility of their decisions to the collective while being protected from its consequences (Gorgeon, 2015).

In the context of projects, this could mean that a manager may be more inclined to take certain risks, knowing that they personally won't be exposed to the full potential cost of a risky decision, instead impacting the organization, or the taxpayers in the case of a governmental organization (Usher et al., 2024).
There is a growing demand for this idea in project planning, where decision makers have more to lose on decisions that were made based upon their information or opinions (Usher et al., 2024).

3.3.3. Considering black swans

This strategy builds on the fact that *black swans*, i.e. high impact events that cannot be predicted, need to be taken into account by managers by building antifragility into the system. Black swan events cannot, by definition, be predicted and identified prior to their occurrence which means that traditional risk management tools won't be sufficient in handling them (Usher et al., 2024).

4 Methods

4.1 Research approach

The overall research philosophy takes inspiration from the principles from Lincoln and Guba (1985) in *Naturalistic inquiry* and the naturalist paradigm, which differs from the positivist paradigm in multiple ways. One being the notion of seeing reality as being constructed in the mind of the individuals, and opposes a view of an objective reality which can be tangible and fragmentable, one part of this is the understanding of e.g. labels and stereotypes, which we use to form the world, making it easier for us to sort and understand it, as well as theories which are constructions of what we perceive in the world (Lincoln & Guba, 1985). One important consequence of this fact is the understanding of the interaction between the investigator and the respondent,

The study design took an emergent approach, as opposed to a pre-planned, that resembled that of a naturalistic study. An emergent design is motivated by the fact that the meaning and interaction of what will be learned is dependent on the investigator and the context, and will first be understood when witnessed (Lincoln & Guba, 1985). The study took form throughout the investigation and data collection and was worked on horizontally rather than vertically.

The study was not fully deductive, because even though a draft of an initial theoretical framework was chosen, the reason wasn't to test the data according to this theory, but to aid the investigation of the data. Initial theoretical frameworks can however be used in abductive approaches to facilitate the exploration of the data in individual cases and which can lead to the development of existing theories (Thomas, 2010). It was also not fully inductive, as inductive studies aim to build theories from observations about singular events (Haig, 2005).

The research took a general inquiry approach. Because of the depth and context-dependent research question a qualitative approach fits the best. It is descriptive and explains the researched phenomena in as much detail as possible.

General inquiry is an alternative approach which fills a methodological space not fully addressed by the common qualitative approaches (Lim, 2025). General inquiry allows the researchers to navigate beyond the limitations of conventional qualitative approaches, with its ability to adapt and create more contextual and detailed insights, along with more adaptability in the research (Lim, 2025). General inquiry offers a valuable approach for research that tries to clear up or illustrate complexities of human behaviour or perspectives (Lim, 2025).

However, studies with this approach may risk “method-slurring”, which describes issues for congruence at different levels of the research framework due to the potential blending of methodologies (Kahlke, 2014). It can however also alleviate issues of congruence by a flexible procedure by building the research around approaches that fit to answer the research question (Kahlke, 2014). Therefore it is important for this kind of approach to be clear about the data collection strategy and the congruence of other parts of the research (Lim, 2025).

4.2 Data collection

The sampling strategy was based on convenience sampling purposive sampling, where invitations were sent out on email to companies that work with IT-development projects in the Stockholm and Uppsala area and where the participants were selected based on their roles and relevance to the study.
The study does not contain strict constraints that hinders the objectivity of the study given these sampling methods, as qualitative studies often use deliberate sampling methods, as opposed to random sampling (Busetto et al., 2020). The type of respondents (see Table 1) were pre-defined to cover different levels and roles within an IT-project, encompassing participants with different experiences and responsibilities, allowing a broader perspective where pieces of information could begin to be put together in a more comprehensive picture about the studied phenomena.

The number of participants were not pre-defined and the data collection process continued until all the relevant perspectives of the studied phenomena were covered. This process was related to what Busetto et al. (2020) suggests for qualitative studies, where the data collection stops when no new relevant information can be found, and where further sampling becomes redundant.

Respondent	**Role**
Respondent #1	Project manager / Business developer
Respondent #2	Consultant manager
Respondent #3	System developer
Respondent #4	Solutions architect
Respondent #5	Consultant manager
Respondent #6	Group leader

Table 1. Research respondents

The data collection took the form of semi-structured interviews. Interviews are one of the best sources of information in caste studies, one reason being that case studies often are about human affairs, which best are reported and interpreted by the well-informed interviewees (Yin, 2014). Interviews give insights both to the facts of a certain phenomenon but also about their opinions and insights on certain occurrences (Yin, 2014). Semi-structured interviews use open-ended questions and an interview guide, which defines the broad area of interest and can be created using previous research and literature and improved and adapted at the start of the data collection (Busetto et al., 2020).

The benefit of using qualitative interviews as opposed with written surveys, is the possibility of unexpected topics to emerge, which also helps with decreasing the researcher bias which can be found in written surveys, where the topics of discussion by definition, is limited to the

researcher's own subjectivity and what is already known by the researcher (Busetto et al., 2020).

The interview guide (see Appendix) was based on questions about the handling of risks and uncertainties in general, as well as questions relating to certain well known risk components of software development risks taken from a study by Ropponen and Lyytinen (2000) where they gathered data from project managers in a total of 1,100 projects.
These questions were asked to help the respondents think of certain risks, which then could be expanded on to discuss how they are being handled or the perceived severity of certain risks.

The specific risks that were presented by Ropponen and Lyytinen (2000) and that were included in the interview guide were: (1) scheduling and timing risks, (2) system functionality risks, (3) subcontracting risks, (4) requirements management risks, (5) resource usage and performance risks, and (6) personnel management risks.

Lastly, questions relating directly to antifragility were included, these questions were based on general concepts from the theoretical framework. These questions were made more tangible for the respondents, as the concepts in themselves are quite abstract to be asked on their own.

The interviews took place either at the office of the participants or in a meeting room booked by the researcher, one of the interviews was conducted online in a video call.

The audio recordings were stored locally by the researcher and later transcribed with help from Klang.AI which is a Swedish service used to transcribe audio recordings. Klang AI was chosen because of its safety focus, encryption and compliance with EU data regulations and protection against extraterritorial laws, such as the American Cloud Act (KlangAI, n.d.).

4.3 Data analysis

The data was analyzed and coded with Atlas.ti which is a software used for qualitative data analysis.

The data was analysed using thematic analysis. Thematic analysis refers to the process of organizing textual data thematically according to codes that may be predefined or that emerges in the data (Cassell, 2015).

The transcribed interviews were firstly read through to get an overview of the data. Secondly, each transcribed interview was coded into the smallest bit of relevant information that can stand by itself, in a process described by (Lincoln & Guba, 1985) as *unitizing*. These bits of segments were often paragraphs or sentences, describing a certain phenomenon in the data. Thirdly, each code was assigned a category, which is a collection of codes that relate to the same content. The process for this categorization was taken from Lincoln & Guba, 1985, where they describe a guiding method for this process. The guiding method begins with choosing the first code as the first category, then you take the second code and decide on tacit or intuitive grounds if this code fits into the first category with the first code, based on if its

content is essentially similar, if it does, you place it in the first category, otherwise the code becomes the second yet to be named category. Then you proceed with this process, placing all of the codes in a category where it fits. The categories are then looked through to see if there exists any overlap in them or if a category needs to be divided into subcategories.

The categories that resulted from the data analysis were based on different types of risks, different general strategies and antifragile concepts and some of the codes were part of several themes. The categories are General risk management, Communication and roles, Crisis management, Resource risks, Scheduling risks, Client/customer risks, Budget risks and Technical risks.

4.4. Research ethics

The research was conducted with ethics in mind before, during and after the data collection and research. The participants rights were communicated in the initial conduct via mail, and further explained before the interview. These rights included (1) the right to remain anonymous, (2) the right to opt out of the study retrospectively and (3) the right to take part of the interview recording and transcription.

Before the study it was made clear to the participants that an audio recording of the interview was needed for the conduction of the study, that the study will be published in DiVA, and that an examiner may take part of the data (i.e. audio recording and transcription).

In the result chapter, data that could be used to identify the participants were left out and exchanged for generic descriptions, for example if they named a certain company or a certain piece of information that could be used to identify them, it was exchanged for "[*company*]" or similarly, in the text.

5 Results

Here, the results from the data collection are presented and structured according to themes found in the data.

Respondent	Role	Time
Respondent #1	Project manager / Business developer	57:11 min
Respondent #2	Consultant manager	61:19 min
Respondent #3	System developer	66:19 min
Respondent #4	Solutions architect	62:10 min
Respondent #5	Consultant manager	61:56 min
Respondent #6	Group leader	52:44 min

Table 2, Interview respondents.

5.1 Results

5.1.1 General risk management

All of the respondents used some kind of risk management method, often based on prior knowledge and experience about working in IT-projects, and also formal methods that are created for project management. Respondent #6, for example, used the PPS-model which includes models for managing risk and classifying risks based on their probability and consequence.
Respondent #5 used a similar approach where risks and problems were defined in colors red, yellow and green, where red meant that they needed external help (with funding etc.), yellow that they could solve it themselves and green where it was going according to plan. These classifications were then updated and discussed continuously by all members of the project.
Respondent #3 worked a lot with safety critical systems where the potential risks were high. Here it was really important to document and test every step of a system integration and think about the worst outcome at each step and how to address it to make sure it doesn't happen.

Risks were however dealt with and analysed in a lot of different ways depending on the project and which people were involved, sometimes the risk analysis was a big part before and throughout the project, and sometimes it was something that was dealt with as it became a problem.
The customers in a project also played a big role in how much resources were spent on risk analysis and the handling of risks throughout the project. Most of the respondents acknowledged that there often exists different goals and motives between the people working on the project and the customer. This has led to that sometimes, current and future risks

haven't been taken seriously by the customer, sometimes leading to big cost overruns or other problems. Many respondents indicate therefore that it is very important to have a continuous dialogue and communication between different stakeholders. It is very important to have the client and developer understand each other, and many respondents highlight the problem of not having the business and the technical side understanding each other.

> Respondent #4:
> "*The biggest risks I have in my organization are lack of communication between the consultant who sits in-house, the client and the developer. I have a lot of developers who have difficulty with this and I also have consultants who have difficulty with this. Where development and business meet, that is the sensitive area. It is often the case that you either come from the operations side or from the development side*."

5.1.2 Communication and roles

Many respondents have mentioned a problem in which the client doesn't fully understand what system they want to have built. It can be that they don't understand how the user would want to use the system or that they don't understand the complexity of a certain implementation.

Respondent #5 talks about how the communication and complexity changes with the size of a project or organization. Respondent #5 describes that it was easier to handle risks when a group was made up of fewer people that knew each other and that the risks and complexity increased when people were spread across the world in different locations. A solution for this was to divide it up into different units that had a certain responsibility, and then to communicate as much as possible between these groups.

Respondent #3 highlights the importance of good communication from the project management to the developers and vice versa, and that early involvement of developers on the planning of the project is a good way to handle uncertainties.

> Respondent #3:
> "*...because you put in the time you put in, and usually the time estimate is made by a project manager, or someone who plans for the big project, and I would say that good communication all the way down and then up again is what's important, because it will take the time it takes, but it's important that you somehow try to convey that this will take the time that the developers think it will take, rather than the time that the project manager had thought it would take. And if you do it really well, you involve developers in the process of planning the time from the beginning, which certainly doesn't always happen*."

Some respondents also talk about being clear and honest about the expectations to the client early on, to lower the risk of a customer not being satisfied with e.g. how long time a project takes. It has also been mentioned that it is really important for transparent communication, where a project manager has the duty to bring up risks to the customer or steering committee, and that the communication is clear and concise.

Respondent #4:
"*That the deadlines that we set, they should be able to be met. It has to be solid. Not taking a chance or aiming for or "oh that would be great". No. We are quite tough with our customers to say that outright. It's not realistic. It will only lead to me having to pull the emergency brake. Their organization will be pissed off, my organization will be pissed off. So if we reschedule for three months later instead, then we know that this will hold*."

Respondent #6 talks about the value of an experienced project manager, who can discover and manage the risks well. A more inexperienced project manager might not be able to handle risks properly in longer projects and when these risks finally emerge, they can create huge problems for the project. Respondent #6 also mentions that sometimes there exists role-clashes or undefined project roles, which creates problems for understanding who is responsible for what, and where risks and problems are not addressed or taken care of correctly, which can lead to significant consequences later on.

Respondent #2 talked about their company having bootcamps for their clients to get together to learn from each other with seminars and presentations to discuss certain topics, and having platforms for them to work and cooperate together, hopefully creating value that Respondent #2's company can benefit from.

5.1.3 Crisis management

The management of crises is also part of the management of risks. Several participants described the impact of Covid-19 on how the business was conducted, and how shifts happened in how people work and how customers invest in projects. Respondent #2 were initially hit hard by the increase in delivery times for hardware components which meant that their customers also were impacted negatively. Throughout this they maintained clear communication and looked into alternative solutions and minimizing damage. In the end the crisis was relatively positive for them, as more companies demanded tools that could allow more remote work.

Respondent #2:
"*It was a crisis for everyone. But otherwise, it has been positive for us when all businesses realized they could work remotely. It still relies on having devices and all the digital stuff working and all those kinds of things. So it has been positive*."

Respondent #4 explained that the pandemic initially caused a stop in the demand, but then that some customers saw it as a good opportunity to focus on investing in the type of technology that Respendent #4 works with, which is explained to be quite safe from external risks, making them able to handle the pandemic better.

Respondent #4:
"*What happened during covid, for example, is that many customers came to a complete stop. Stopped projects, development, stop. Then we had other customers who thought, finally we have time to spend some money and time on our business system. And our*

industry is part of the IT industry which is old. It is not that hip to work with business systems. People want to build apps and work with AI and all that. But we are also very stable in the way that all companies have to have a business system. This is a must. This is an infrastructure that all customers need to have. So we don't feel that exposed to risk from the outside world. But this is industry-specific."

Most respondents said that there were some initial problems but that overall the companies did well, maybe because of the healthcare sector which is big in Sweden and Uppsala in general.

Respondent #6:
"*One realized that it was necessary, partly that it works well to work remotely, which they are now changing back. And that, I think productivity increased and the focus was great. I guess many of the products we develop in Sweden were needed during covid*."

Respondent #3:
"*For us, it was actually positive, because the pharmaceutical industry in Uppsala really flourished, because of all the different tests and reagents and all that stuff produced here, there was a need for more of that, so we had a lot of work, and it's a large customer base."*

In general the effects of the pandemic were described as positive, most of the respondents saw an increase in productivity, with more flexible work schedules and new ways of working. There however exist nuances with for example Respondent #1 describing some problems that of course can arise when more people begin to work from home, and that in the end, it is up to the employer to make sure it works as well as possible.

Respondent #1:
"*I think it's good that there is flexibility and that one can utilize that way of working. Then I believe there is a lot to gain from it, of course. Because in flexibility, everyone has a daily life, there are puzzles to solve, you might have children, you might have pets. You should manage many things and it becomes a less stressful time, day, and environment. More flexibility. But something is lost. What I think is lost is the commitment, the team spirit, and perhaps ultimately the quality in the deliveries that the team is supposed to produce. It is primarily up to the employer or the client to set those demands and follow up on them*."

5.1.4 Resource risks

A common risk is that people that are expected to be involved in the project may not be available when they are needed, or not as available as expected.

Respondent #1 explains this problem when working with organizations where SAFe (Scaled Agile Framework) have increasingly become the norm. SAFe makes the organization more cross-functional, where people work on activities which are prioritized by the organization rather than dedicated to certain projects and even though he sees the benefits with it, it can

become a problem when he as a project manager needs to plan for which resources that can be expected.

> Respondent #1:
> "*From a project perspective, if I come in as a project manager somewhere, and it's a type of organization where you work a bit hybrid, then it's deeply unsatisfactory for me that I can't be allocated resources with certainty, to say that "Stina" works 50% on this project, without her being available depending on what's most important based on a priority. On the whole, it's a bit of a problem for me. But if you work purely, it's maybe a bit easier*."

Respondent #6 also discusses SAfe, where the group is split into several "delivery teams" and where every team member is part of two to three teams. The respondent discusses that this normally isn't a problem when it comes to IT maintenance and operations, but when it comes to projects, the respondent would like to have full time resources, which however can be hard because the group has many different tasks.

Respondent #5 discusses Sweden's new employment law which impacts firms hiring consultants to work under their own control. This law states that hired consultants must be offered a job at the company, with fair pay according to the labor union, if they have worked more than 24 months at the hiring company. He explains that a lot of larger companies have a hiring freeze so that they can't offer employment, leading to them having to lose the consultant and hiring a new one. This creates a problem for both the consultant firm, as they lose a deal, and also the hiring company, as they lose a valuable resource. Respondent #5 explains that, in IT-projects, the necessary skills and the needed domain-knowledge, takes a long time to develop, as the requirements are seldom clear and the complexity high.
This has lead to that respondent #5's organization has lost deals, which have led to them shifting more towards service-oriented contracts where they will work under their own control on a certain part of the project.

Respondent #2 discusses the uncertainty that can occur when they want to lease a consultant to a client for a job in the future and the availability of this consultant is not certain. They then have to weigh this risk together with the risk that the client might go to another company and that they lose the deal.

Another risk is that knowledge can get concentrated on certain people, or that progress potentially can get lost if someone is sick or if a computer breaks. This highlights the importance of having some sort of redundancy in resources, where someone can take over some work activities if the main person is sick or leaves the organization, or where the code base is frequently being updated - protecting against hardware failures etc.

> Respondent #3:
> "*We work with feature management and git. A big part of the reason that you push things every day is so that everyone has access to everything. So that there is no one sitting at home building something and then when they are sick, half the code is gone. Or a computer breaks or whatever it may be. But then it is the case that there is expertise that is very important and if it stops, it will be difficult. It is a big risk for*

companies in general and for projects. And it is something that you try to get rid of. So that usually such people need to spend quite a lot of time teaching and that you "go double" with that type of person. Because there are definitely projects where there is someone who is sitting and is very important. But it is a huge risk for the company and that person can more or less ask for whatever salary they want in the end. So that it is something that you really want to work away from in one way or another."

This risk is shared by many respondents, which is why in many cases, the organization tries to eliminate or decrease this dependency by having multiple people working on the same system or code base, or having senior employees supporting and overseeing junior colleagues.
Respondent #5 discusses that in small projects, one solution can be that there could be two developers working 50%, decreasing the technical risk by pair-programming or having someone to discuss with, making the code more readable.

Respondent #2 explains that they want to deliver in teams as much as possible, to keep the expertise in the project in case one person leaves, this also makes it easier for them to take in the personnel in the project. Respondent #6 shares this view and explains that they have a work environment goal that no one should be alone with a system and that at least two people share crucial knowledge.

Respondent #5 describe that they have a "Linux group" at their organization that have monthly meetings where they can get together and work on different projects, either private or work related, as a way to innovate and gain knowledge without spending much resources.

There can also be redundancy in what work activities the people working in projects are active in. Respondent #4 explains that the people are not fully occupied with planned activities, instead there exists a redundancy and extra capacity where unexpected events can be taken care of, without impacting the main work activities.

Respondent #4:
"*We live in a world where "missiles" come into your inbox from right and left. And then the project participants have the opportunity to fire the missiles. In our case it is that you fire them over to supporters. Or if it comes in from other customers or you fire it over to colleagues. That not everyone in the organization is busy with projects all the time, but there are some who go and tinker a little with administrative matters that can take a missile from a colleague*."

5.1.5 Scheduling risks

Some respondents mentioned the hardship in balancing deadlines, budget and quality of the system, and that it is often the case that

Several respondents highlighted the difficulty in estimating how long a project will take. Several causes for this were presented, such as the lack of information that was available at the start of the project, the project complexity, lack of communication between different parts of the project or that there exists a bottleneck somewhere causing delays.

Respondent #6:
"*The time estimate is often the big issue. And that's where it's really hard, because the only thing you know about the time estimates is that they're wrong*."

Respondent #5 described the lack of communication in one case where input from a team member was ignored before the initiation of the project, which resulted in huge delays.

Respondent #5:
"*I remember once when one of my architects said that "I don't believe in this. It's just not going to work at this time". Then I said that I won't sign the paper as development manager. Then my boss signed it instead… But I can say that that project didn't make it. It took three times as much time as they had estimated. And then it became "too big to fail". So then it was just a matter of continuing to work."*

Some solutions and heuristics were presented that involved the project arrangement, communication, structure and iteration.

Many respondents noted that clear communication and follow-ups from the developers up to the project manager is important, and also to involve the developers early on in the time planning process.

Respondent #4:
"*And then follow-up all the time. Never let go of follow-ups. The project manager has to be on people and when they don't bring their input into the project, you have to get tougher on this. And if they haven't done it in a few weeks, I say okay, it has to be discussed with the companies. If you don't bring your inputs, we can't go live. So very clear. This is just general project management. You need to be very communicative and you can't be tolerant. You can't turn a blind eye when something doesn't go right all the time. So it's also something that we inherit from us old people to the younger ones, to be a little tougher in project management.*."

This also ties into managing expectations, not being too optimistic, which also is an important point that has been mentioned, because there is very much that cannot be predicted.

Respondent #5:
"*That's the absolute biggest problem, I think, that all companies have, not just in IT. It's that you have tons of projects, you have a management that wants everything to be done at the same time. Then they get like this, "but prioritize this". And then everything has top priority. But everything can't have top priority. If everything has to be done at the same time, it's like you have to do 20 things in an hour, it's 3 minutes per thing since you set a time. That's probably the absolute biggest problem, that those who order this, those who decide, they can't prioritize*."

Respondent #6 also mentions taking everything into consideration that can affect the time for the projects, meaning not only the time for the actual development, but meetings, testing and

audits. This also strengthens the view that there needs to be several stakeholders involved in this process.

Respondent #4 talks about breaking down plans and projects into smaller pieces and not planning too big and too long into the future. He explains that it is very important to be able to deliver, both for the project team and client, because everyone learns something along the way, which you can use later when devising a new plan.

Respondent #3 explained that it's good to add 30% to the expected time of the project by default, but then also to set time goals incrementally, as it can be dangerous to have deadlines too long away in the future, as you think you have a lot of time left.

Respondent #4 also explains the benefits of working incrementally, learning as you go and adapting to new circumstances and knowledge.

> Respondent #4:
> "*So it's also about taking things apart. You might not do everything to the final scope. You go to scope 1, go live. Then you've learned a lot of things that take a new scope, go on a new project, go to the next step instead of wanting to go all the way in a project. So for us it's very important to deliver. Because the customers, the developers, everyone learns something along the way. So even if you're planning something over here, when you get here you actually know more. So pause in between*."

Respondent #6 mentions a heuristic where you take the average of three possible scenarios for time estimation, the best case, the expected case and the worst case, this gives the perspective that the estimated time is on a span between different possibilities and might skew towards depending on the extremes.

5.1.6 Client/customer risks

Many of the respondents mention the risk in that the customer of the project does not fully understand what they want being built, that they don't understand the complexity or that the system did not take into consideration the end users of the system. Many times a system could be built according to the set out requirements, but that it needs to be redesigned because it actually didn't solve the problem that the client wanted.

> Respondent #2:
> "*It may be that "we know our business is covered by this. We know we will need help but we have no idea how." Then they can say more like this, "Please [company name], can you help us based on the organization we are? Can you come back to us with a proposal?" Yes, we can. It is almost better that we get to formulate a proposal and get back to them. We think like this and you can have a discussion based on that. But sometimes, and that's often the case anyway, they ask us the question :"We need some help with this" But sometimes it can really be that they want a square Christmas tree. "Okay, but are you guys really sure about this?" "Yes, we are". Okay, but you are the one paying. We write an agreement that you want a square Christmas tree and then we*

will fix a square Christmas tree. And very often, it can also be that they get quite a lot of counter-questions from us. "Have you considered this and this?" "Yes, we have". And then when the work gets started they say, "this was wrong". "Yes, it was, we knew that too". So it happens occasionally. But we often talk about challenging our customers. So we really want to do it in the way we think is best based on our experiences and what the customer is doing."

Respondent #2 mentions that Proof of Concepts can help with proving a solution for the customer or finding out the needs of the customer on a smaller scale. For example they can design and implement a solution on an isolated part of the organization to explore the viability before implementing it company-wide.

Some respondents suggest that having a transparent dialogue with the customer is very important, especially in the beginning, when discussing the system. Another point that has been mentioned, is the importance of discussing with the client about what problem a certain solution wants to solve, and not just the artefact itself.

Respondent #4:
"*[Clients] can often think that things are very simple and that you have to be able to do it in three months. So that's a very typical problem I have. I need to explain to our clients what complexities there really are in such projects ... In the sales phase, it's extremely important for us to, like, we want to sell ourselves, of course, but we also want to tell the customer what.. through an assessment, what's easy, what's difficult? Should we solve things technically or should we solve things through management decisions*?"

5.1.7 Budget risks

Another risk is that of estimating a budget which many have described as very hard, given uncertainties and complexity that isn't clear at the start of the project. A big risk is also that the costs can escalate very fast since IT projects in themselves are very expensive.

Respondent #1:
"*IT development costs a lot and businesses don't always understand that. If we take an amount of 300-400,000 SEK in the context of the business, it's a huge amount of money. But in IT, it disappears quickly*."

Another risk which has been mentioned by two respondents, is that some projects are being started carelessly even though there is a lot of uncertainty about the costs and financing, and if the costs were to be known in the beginning, the projects wouldn't have been started.

Respondent #5:
"*I think if you find out what it would really cost. Then maybe you wouldn't even start a project. So sometimes I think you don't want to know what it's going to cost. And then*

when you've run out of that money, it's just, "now we've invested this much, we can not NOT continue". I have many examples of this."

One important point that has been mentioned to handle these budget risks is to have a good dialogue with the steering committee and that it is the project manager's responsibility to communicate expectations to the steering committee, especially the client.

One suggestion is to have a "risk pool" of money that is around 20% of the project's budget, only made to handle unexpected events that can occur. In respondent #6's case this pool of money often gets used up quite fast, but the purpose is only to use it in cases where it really is needed, and that has a big impact on the project.

Handling these budget risks are also noted as being very hard due to complexities of the project, and you can often handle these risks, although at the expense of quality of the system, such as getting a legacy code base that will be hard to maintain and support in the future.

5.1.8 Technical risks

There are also technical risks that can appear in the IT system, either by itself or as a product of the general uncertainty and uncertainties in the project overall. Technical problems have been described by multiple respondents, as hard to detect and to predict, especially by respondent #3 that mentioned the complexity in IT systems that involve hardware programming. Respondent #3 noted that memory leaks are the most troublesome problem and risk as a programmer of hardware systems, and that the cause of this problem is often unclear and seemingly random. Memory leaks are basically the phenomena of when a program uses up too much memory over time, gradually leading to the computer slowing down and sometimes eventually crashing. Respondent #3 noted that these problems can become extremely expensive, as they are hard to localize and understand.

Respondent #3:
"*Yes, what can happen is the absolute most difficult thing I've ever experienced and that's memory leaks. Finding memory leaks is difficult. These are things that are very difficult to test and very, very difficult to find the fault if it's not obvious. And then the costs can also run up considerably*."

One way to handle this, at least partially, that has been described is to have a system that as soon as possible can find these errors, so that you can move forward faster. It has also been mentioned that it is important to have the logging of events that happen in the program, so that you can know what process or part of the program is causing the errors.

Respondent #4 suggests that especially when building big artefacts from the ground up, you should build as little as possible yourself, since a lot of solutions already exist, and have been built before.

Respondent #4:

"*In any case, when you build such big things on procurement from scratch. Ideally, you should build as little as possible yourself. Most things are available. Also good tips for everyone. Most things have been built at some point. "So don't try it again". Then there are innovative things. You want to become a world-leading company in search engines. Then you have to build things that no one has built before. Otherwise you will never get rich. Most projects are not like that*."

In general, has documentation, testing, and traceability been suggested as a good way to handle risks and ensure software quality in the system functionality. However, these steps often increase the need for resources and increase the time of the project. Respondent #3 explains that they have 100% test coverage and extensive integration testing in some projects, but that this depends upon how critical the functionality of the system is, indicating that more resources are spent on mitigating errors that have potentially more severe impacts.

Respondent #5 mentions that for safety critical systems, two codebases could be developed separately for the same thing by different people and before they are deployed, the two codebases get tested and the best version is the one that gets used and deployed.

One risk which Respondent #3 and #5 mentioned is that sometimes the user interface isn't properly evaluated and designed for the users, which introduces a risk where the users may not use the system or application in the way it was intended. It is important to get user feedback on an application and it was suggested that having an UX-designer test the system could mitigate this, even though it often is not prioritized.

Another technical risk that have been mentioned is by Respondent #4 which is the reliance on another part for a platform they customize and deliver to their customers. This platform has a legacy problem and large complexity and often comes with bugs with new updates, and especially because when the updates are released, they are deployed to all customers directly, as opposed to before when they were gradually installed locally to every customer, making the errors less impactful.
To handle this risk Respondent #4's company has begun to decrease the amount of technical customization to this platform, as these customizations often collide with the updates from the platform. To handle this vulnerability from a 3rd party they also have good agreements and communication with their customers, being transparent about the expectations, and keeping the customers up to date.

Respondent #4:
"*We make an agreement with our customers that we don't take responsibility for anything. It's always [the platform's] fault. It doesn't exactly say that, but that's basically it. We make no guarantees whatsoever … Then all customers understand that sometimes something goes wrong. But what has happened is that we have started to reduce our own build. Because we have noticed that our code, when [platform] changes its code so much and our code connects with [the platform's] code. Then our code can stop working when they change things. And if you have too much code out twice a year, you will have hell getting things to stay after an upgrade. So we are strategically reducing our own build of code. So we are much more selective in what we do. We tell the customer much more often that it is not worth it for us to build this away. Wait a year and maybe [the platform] will come up with a feature that does*

something similar. So we definitely reduce our own vulnerability in the form of code. In reaction to [the platform] being able to release missiles when they screw something up."

5.2 Analysis

The analysis chapter will look at the results from the antifragile concepts described by Usher et al. (2024) to see how the practical risk-handling methods and strategies used by the respondents are looked at through the theoretical framework.

5.2.1 Simplicity

Subtraction

Subtraction which is a subconcept of simplicity means to remove unnecessary elements to make processes and decision making simpler. Eliminating unnecessary components and focusing on the core-components makes it easier for organizations to handle risks and to adapt to changes.

Subtraction can be seen, when talking about technical risks about not overcomplicating solutions, in many cases, certain solutions already exists, so you can reduce the technical uncertainty and complexity by not trying to build something that already exists.
It was also noted by another participant, that they use technologies from known vendors, and that the trust in these products already was established, so that the company only needed to make customer-adapted solutions in some cases.

A lack of subtraction was also identified, where a participant described that in one project, they had three different front end frameworks for their development, which introduced a risk where the complexity became bigger, and where three different systems needed to be kept track on.

There were also some times when there wasn't a solution or understanding of complexity, for example in cases where different customers were not interested in thorough documentation, which would make the system easier to understand, or when there was some legacy system written 25 years ago without any documentation.

Less is more

The subconcept Less is more emphasizes that simpler solutions are less complex and more flexible, and that heuristics can be used favourably when faced with uncertainties.

Less is more could be seen in a few cases. For example when discussing the handling of scheduling risks and time estimation, some respondents mentioned that the use of simple

heuristics is the preferred method, such as adding 30% to the estimated time plan, or creating it based on an average from best case, probable case, and worst case, or simply just inviting a pessimistic person you know to the meeting.

Agile methods can also be attributed to the Less is more concept, where many respondents preferred to work iteratively without an overreliance on detailed plans. This helped the respondents in being flexible to new information, and where they could respond to problems and risks faster.

5.2.2 Development strategies

Building optionality

The subconcept Building optionality refers to creating options and flexibilities to be able to handle risks or uncertainty. It refers to having multiple alternatives where you can adapt to new situations or requirements and having the possibility to choose alternative paths. It builds on both redundancy, which means having resources to be able to make better decisions, and trial and error, which means experimenting without much stake, to gather as much information as possible.

Many of the respondents mention strategies that are related to optionality, where you either experiment in some way, or use techniques that with low stakes, aim to gather more information about a certain area. One strategy which many describe as important, is the use of a pre-study when gathering information about a project, making it easier to understand the complexity. This can aid in making better decisions about the necessary budget, the planned time necessary and the expected system quality and performance. All of this helps the client and developer to manage expectations, and to make it clearer about what is realistic. Another, although extreme, strategy that is mentioned is that companies which are very innovative may begin several projects that they run in parallel and then later choosing the best one, and abandoning the others.

Several organizations also use Proof Of Concepts as a low stake option to explore a certain idea, without going all in from the beginning, leaving doors open for different decisions, increasing optionality.

There have also been discussions about redundancy in work activities, for example where two people work on the same task, which means that you have the ability to handle unexpected events, without negatively impacting the planned main work activities.

There was one example from the public sector of when projects sometimes were started without proper information about the financing, which would decrease the optionality and increase the uncertainty of if a project could be completed.

Another example is a company that built a lot of their systems on-top of existing software, which causes a lot of dependencies and minimizes their ability to adapt to changes.

Modularity

Modularity refers to breaking down bigger, more complex systems, into smaller more manageable pieces. This makes it easier to update or modify certain parts of the system, without impacting the whole system.

Agile methods, which is something many of the respondents use, is also an example of modularity, where you break down a plan or goal, in smaller iterations, which are simpler and less complex than planning the whole project from start. There was however one problem with the view that the agile way of working was more of a formality and where in practice you wouldn't do it in an agile way.

Another example is the splitting up of deliveries of a project into smaller more manageable parts. For example where sub-systems can be integrated one-by-one instead of all at once. This helps with the adaptation of a new system and can avoid big failures. This point also matches the trial and error strategy, where you get valuable information about the feasibility for example, while taking a smaller risk.

5.2.3 Risk and opportunity strategies

Barbell strategy

The Barbell strategy means that you invest a majority of resources in safe alternatives while taking high risk with a smaller part of the resources. This strategy tries to minimize the risks and potential losses, while still being able to benefit from potential wins.

The Barbell strategy could explicitly be seen in a case, where the organization put together a group of experts on Linux that could get together once a month, to discuss and work together on their projects, similar to the idea from Google, where their employees spent 20% of their time working on experimental projects. This strategy does not have any big downside, as it only happens once per month, but can have big benefits for the organization by gathering experts and having them work on innovative projects.

Another example, related to the Barbell strategy, is that one organization has decided to invest resources into bootcamps and learning platforms for collaborators and customers, where new technologies and challenges could be discussed, in hopes of reaping the rewards of this program, by getting valuable knowledge and increasing their relationship with similar companies. This was discussed as a success factor for them as a company, since it gathers a lot of smart individuals, benefitting from the knowledge and solutions that can be presented, leaving a lot of upside available, while still keeping the downside low.

Skin in the game

Skin in the game refers to the idea that decision makers also must be exposed to the risk that comes from the decisions that they make. It increases personal responsibility and lowers inappropriate risk taking. It means that the people involved have something to win or to lose and lowers the risk of failure.

When talking about personal responsibility, most of the respondents had a nuanced view on how it took place inside a project with all the team members, but that there sometimes could be a clash between the goals of the steering committee or the customer, and the project group, where there were sometimes discussions about who would take the blame if it failed.

Considering black swans

Considering black swans is about preparing for extreme and unpredictable events. This idea encompasses all of the above and can help organizations plan and acknowledge that it can happen.

This idea connects to for example the Covid-19 crisis, where most of the respondents considered that they handled it well, not because of the prediction or knowledge of the event, but the fact that they could handle it, and in some cases come out of it with a comparative advantage.

There also existed a case when one respondent described the event when a consultant disappeared from the project with all of the code of an application, which was something that they couldn't prepare for and handle, which showed a negative consequence of something unpredictable.

6 Discussion

This study has given insights into how risks and uncertainties can be dealt with in IT-projects and how it relates to the concept of antifragility. The study used a framework by Usher et al. (2024) for antifragility in projects and where the term antifragile was originally defined by Nassim Taleb, as a way to describe certain systems that have the possibility to get better with stress. The framework was used to see in which way real world efforts by the respondents and their experiences relate to antifragility.

The results present the biggest and most common risks that IT projects are prone to and how they may be handled. The findings show the importance of transparent communication between different stakeholders and parts of the project team and how different unaligned goals can create risks for the project. It also shows the importance of simplifying processes and to use simple heuristics for minimizing the complexity.

The study has given insights into how the handling of risks can be looked at through another theoretical lens, that may explain why certain efforts work and why others don't. What has been shown is that a lot of respondents use successful strategies related to what is described in the framework and that there may arise problems when these strategies or concepts are overlooked.

Because of the diverse background for the respondents, considering they had different roles within a project, the data that could be gathered could fill in different gaps from different respondents, creating a more complete description of all the risks and problems and in which way they thought that they could be dealt with.

6.1 Limitations

Only six respondents were interviewed, which gives limited insight to thoroughly examine and analyse how different roles and companies think about and handle risk. Because of the quite broad background of the respondents, the study had a broad overview of the area, but it would be hard to do any deeper systematic analysis on how these different roles experience and handle risks differently.

The study data collection only took place in Sweden, which impacts what broader insights this can mean for other places. Projects in general can work very differently depending on which country you are in, for example different hierarchies and communication styles, as well as which sectors in the country that are important or the economic capabilities.

Another limitation is the difficulty in getting the honest opinion out of a respondent, it can be hard as an employee of a company to criticize certain actions from the company or how different people have acted. Subjects about how the company conducts business can also be sensitive and the respondent might not want to make themselves or the company look bad.

6.2 Recommendations for future research

Future research could look into other areas that may be interesting to study from the perspective of antifragility. The concept in itself is very general which makes it suitable to apply to other areas as well, for example other types of projects, or other parts of software development. Other studies could look on a more technical level as well, such as software testing or different technologies, such as cloud or AI.

6.3 Conclusions

The results show that antifragility could be an interesting concept when analysing risks and uncertainties in IT-projects and information system development. Many heuristics and processes used by the participants in the risk management of IT-projects can be analysed and have their ground in antifragile concepts.

Risk management in general is hard and complex but there are ways of how it can be managed. A lot of the data concerns the importance of communication, flexibility and to be able to plan continuously and adapt. A lot of risks concern how resources, budget and schedule are managed, which all have a strong part in the managing of a project.

This study presented a new way on how to look at risks and may help project managers to better handle risks and uncertainties in the future.

7 List of references

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Appendix

Interview guide (swedish)

Min studie:

- Jag undersöker hur risker och osäkerheter kan hanteras i IT-projekt
- En fallstudie
- En kvalitativ studie som genomförs med intervjuer

Rättigheter och villkor

- Jag kommer att spela in ljud som kommer att lagras lokalt på min enhet.
- Uppsatsen kommer att publiceras i diva-portal.
- Deltagandet kommer att vara anonymt.
- Examinator kommer eventuellt att få tillgång till rådatan (inspelning och transkribering).
- Du har ska ha rätt att dra tillbaka sitt deltagande i efterhand om du ångrar dig.

Enklare bakgrundsfrågor

- Vad är din roll på företaget?
- - Vad arbetar du med på daglig basis?
- Hur länge har du jobbat med det du gör?

Huvudfrågor (störning/risk/osäkerhet/problem)

- Vilka är de vanligaste riskerna med det du arbetar med (systemutvecklingsprocessen (mjukvara, processer, applikationer?)
- - - skriv ner dom. sedan fråga hur var risk hanteras
- - - Vad tror du är ett bra sätt att hantera dessa risker på?
- - - största risken? hur hanterades?

- Använder ni strategier för att hantera risker?
- - - skriv ner dom, (koppla till om dom gynnas missgynnas, tar tillvara på risker)

- Har ni några standardiserade / generella riskhanteringsmetoder?

- Har du upplevt eller hört någon kris på företaget?
- - - Hur hanterades den?
- - - var den förutsägbar?
- - - hur diskuterades det efteråt?

- Har ni något sätt att förhindra kriser på?
- Hur tror du man hanterar kriser på bäst sätt?

- Hur fungerar planeringsprocessen (planeras det mycket på förhand?)

- **Fråga om (och hur de hanterar):**
- Schemaläggningsrisker
- Systemfunktionalitetsrisker
- Underleverantörsrisker
- Kravhanteringsrisker
- Resursanvändning- och prestandarisker
- Personalhanteringsrisker

antifrag.

- Vad är den generella inställningen till utmaningar?
- Har individen mycket beslutsfattande?
- Skulle du säga att det finns flexibilitet i besluten ni tar?
- Finns det flera sätt att lösa ett problem på?
- Hur ser du på experimentering (prototyper osv)?
- Hur ser du på personligt ansvar?
- Hur ser du på lärande?
- Har ni krävt genomföra en stor ändring på företaget (pga någonting?)
- Vad tror du är anledningen till att projekt blir lyckade och/eller att ni har överlevt och frodats som organisation?

Sista fråga: Är det något som du trodde eller ville att jag skulle ta upp som jag inte har gjort?

Avslutande

Om du är intresserad så kan jag skicka transkriberingen till dig om du vill säkerställa att det du har framställts på rätt sätt.
Har du någon avslutande fråga eller tillägg?
Tacka för deltagandet.